Research Statement

Diego R. Känzig, Northwestern University, October 2026

I am a macroeconomist studying how major economic, policy, and environmental shocks shape growth, inflation, welfare, and inequality. My work asks how climate change and the low-carbon transition affect economic activity and welfare; how disruptions in energy markets, supply chains, and trade propagate through the economy; and how heterogeneity and expectations shape macroeconomic dynamics and policy transmission.

A unifying theme of my work is identifying aggregate effects. Cross-sectional evidence captures differential impacts across households, firms, or regions while absorbing common effects—the 'missing intercept.' I address this challenge using time-series evidence drawn from financial markets, narrative and historical records, and natural climate variation. My research combines this aggregate evidence with micro data and structural models to uncover mechanisms, quantify welfare, and conduct counterfactual analyses.

Climate change and policy. My first line of work studies the physical and policy consequences of climate change, bringing together new data, empirical analysis, and structural modeling to quantify the associated economic risks and policy trade-offs.

Measuring climate damages. A central question is how to measure the aggregate economic damages caused by a warming climate. Much of the empirical climate literature exploits local temperature variation across countries or regions. This provides credible cross-sectional identification, but it also removes common climatic variation and may miss damages that operate at a global scale.

In "The Macroeconomic Impact of Climate Change: Global vs. Local Temperature" [5] (with Adrien Bilal, *Quarterly Journal of Economics*, 2026), we propose a new approach to estimating the economic damages from climate change. Rather than exploiting local temperature variation across countries, we use time-series variation in global mean temperature, drawing on natural climate variability associated with phenomena such as El Niño, volcanic eruptions, and solar cycles. This approach allows us to capture the broader consequences of warming the global climate system, including ocean warming and the associated rise in climatic extremes. We find persistent economic losses that operate through declines in investment, capital, and productivity, with damages distributed more uniformly across countries than estimates based on local temperature variation suggest.

To quantify the long-run implications, we estimate structural damage functions in a growth model by matching the model's responses to a temperature shock to those observed in the data. The model allows adaptation through adjustments in capital. As in earlier work, projecting the effects of sustained warming requires extrapolation from moderate historical temperature fluctuations that eventually fade, although we find little evidence of nonlinearities within the observed range. Our estimates imply that a permanent 1°C increase in global temperature lowers world GDP by more than 20 percent in the long run. Additional warming of 2°C by 2100 generates a present-value welfare loss exceeding 30 percent, and the implied social cost of carbon exceeds $1,200 per ton of CO_2.

The consequences of climate change extend well beyond its effects on economic activity. In "Global Temperature and Global Mortality" [16] (with Adrien Bilal and Krzysztof Lisiecki, 2026), we study the impact of warming on human health. We construct a new dataset covering most of the world's population since 1960 and estimate the dynamic effects of global temperature shocks. We

find that a permanent 1°C increase in global temperature raises annual mortality by about 2.5 deaths per 1,000 people, an order of magnitude more than the effect implied by local heat exposure. Much of the mortality response operates indirectly through the economy: global income losses account for about two-thirds of the cumulative effect. Economic contractions can worsen health, for example by reducing access to healthcare, medicines, and nutrition—a channel that we explore in a companion paper "Growth Spillovers and Life Expectancy" [17] (with Adrien Bilal and Krzysztof Lisiecki, 2026).

We embed these estimates in a growth model in which households can respond to climate-related health risks through health spending, while economic growth itself improves health. Mortality consequently adds about \$400 per ton of CO_2 to the social cost of carbon, roughly one quarter of total climate damages. The model implies that private adaptation through additional health spending is relatively limited, whereas continued economic growth substantially cushions the mortality effects of climate change. This creates a double dividend from economic adaptation: policies that make the economy more resilient to climate change also reduce climate-related mortality.

In ongoing work, I aim to provide an even broader assessment of the macroeconomic and welfare consequences of climate change by studying how global temperature affects demand, trade, and financial stability. In "Prices and Quantities in a Warming World" [26] (with Adrien Bilal), we use the joint responses of output and prices to global temperature shocks to distinguish supply from demand effects of climate change. This distinction matters for welfare because higher temperatures can make activities such as travel, leisure, and dining out less enjoyable, lowering the marginal utility of consumption and reducing demand. The resulting welfare losses extend beyond those captured by the decline in measured output. In "Climate Damages Across the Globe" [25] (with Adrien Bilal), we study how climate shocks propagate across countries through international trade. We first estimate these cross-border spillovers, asking how climatic disturbances in trading partners affect domestic economic activity. We then use the empirical evidence to discipline a multi-country climate–economy model and study adaptation through trade. Finally, in "Climate Change and Financial Stability: 1850–2024" [24] (with Adrien Bilal, Emirhan Ilhan, Karsten Müller, and Chenzi Xu), we assemble a long historical dataset to study whether physical climate shocks raise the likelihood of sovereign debt and banking crises. The long time dimension allows us to exploit substantial climatic and institutional variation and to study whether financial vulnerability amplifies the effects of warming.

The transmission and incidence of climate policy. A second central question in climate macroeconomics is how policies aimed at decarbonizing the economy affect aggregate economic activity and who ultimately bears their costs. In "The Unequal Economic Consequences of Carbon Pricing" [1] (*American Economic Review*, forthcoming), I develop a novel identification strategy to estimate the aggregate and distributional effects of carbon pricing. Exploiting institutional features of the European carbon market and high-frequency data around policy announcements, I isolate unexpected regulatory shocks to carbon prices and trace their effects on emissions, output, inflation, and inequality. I find that tighter carbon policies raise energy prices, reduce emissions, and spur green innovation, but also depress economic activity and disproportionately burden lower-income households. These distributional effects arise not only because poorer households spend more on energy, but also because they experience larger income losses through employment and wage channels. These indirect, general-equilibrium effects play an important role in the transmission of carbon policy.

Quantitatively, I use the estimated responses to infer an aggregate marginal abatement cost of slightly above €100 per ton of CO_2. The findings highlight a key policy trade-off: while carbon

pricing is effective in reducing emissions, accompanying fiscal measures may be needed to mitigate short-run costs and maintain public support for climate policy.

In "Carbon Pricing and Inequality: A Normative Perspective" [4] (with Saki Bigio, Pablo Sánchez, and Conor Walsh, *Economic Journal*, 2026), we map out the welfare implications of carbon policy shocks. We estimate money-metric welfare losses by tracing how carbon pricing changes households' budgets through consumption prices, labor income, financial wealth, and government transfers. We document considerable welfare impacts, with losses concentrated among younger, poorer, and less-educated households and in Southern and Eastern Europe.

In "Carbon Pricing and Inflation Expectations" [3] (with Michael Bauer and Glenn Rudebusch, *The Econometrics Journal*, forthcoming), we study how carbon pricing affects inflation expectations and show that carbon policy shocks raise market-based inflation expectations persistently, with effects extending up to ten years out. In "Climate Policy and the Economy: Evidence from Europe's Carbon Pricing Initiatives" [7] (with Maximilian Konradt, *IMF Economic Review*, 2024), we compare the effects of the European carbon market with national carbon taxes. While both policies reduce emissions, the carbon market has larger macroeconomic effects, reflecting stronger energy-price pass-through, differences in sectoral coverage and revenue recycling, a more pronounced monetary policy response, and greater scope for carbon leakage under national policies. Leakage is not confined to national carbon taxes. In "Carbon Leakage to Developing Countries" [22] (with Julian Marenz and Marcel Olbert, 2024), we show that higher prices in the European carbon market shift production and emissions toward the African subsidiaries of European multinationals. In ongoing work on "Green Fiscal Multipliers" [23] (with Lint Barrage), we study the macroeconomic and environmental effects of public investment and subsidies for the green transition and compare their cost-effectiveness with carbon pricing policies.

Beyond the effects of realized policy changes, uncertainty about the future path of climate policy has become increasingly salient amid recent policy reversals and shifts in decarbonization commitments. In "The Macroeconomic Effects of Climate Policy Uncertainty" [18] (with Konstantinos Gavriilidis, Ramya Raghavan, and James H. Stock, 2026), we develop a novel newspaper-based index to measure uncertainty about future climate policy and propose a new narrative instrument to identify exogenous policy uncertainty shocks. We find that higher climate policy uncertainty reduces output and investment while increasing inflation—suggesting that such shocks act as supply rather than demand shocks. These shocks depress activity through lower firm investment and R&D, particularly among firms with high climate exposure. The results imply that a clear and predictable climate policy path is crucial to minimize transition costs.

Finally, in "Innovation, Business Cycles, and the Climate Transition" [20] (with Maximilian Konradt, Lixing Wang, and Donghai Zhang, 2026), we study how the business cycle shapes the pace and direction of innovation. While non-green innovation is procyclical, green innovation is countercyclical. We explain this divergence in a model in which green patents derive more of their value from distant profits, making them less sensitive to short-run fluctuations. In downturns, weaker non-green innovation also lowers the demand for scientists and hence the cost of green R&D, further encouraging green innovation. Evidence from market-implied patent values and scientist earnings supports these mechanisms. An important policy implication is that innovation subsidies are state-dependent: a dollar of R&D subsidies generates more innovation, and relatively more green innovation, during downturns.

Long-run growth and optimal climate policy. In "Balancing Growth in a Warming World" [15] (with Adrien Bilal and Thibault Ingrand, 2026), we take a long-run perspective and study theoretically how the severity of climate damages shapes optimal climate policy in a world where warming continues into the foreseeable future. We develop an integrated assessment framework in which emissions permanently raise the carbon stock. Balanced growth remains possible under climate change, but at a lower rate. The theory reveals two policy regimes. When damages are modest, the optimal and laissez-faire economies share the same long-run growth rate, and a constant carbon price balances mitigation costs against damages. When damages are severe, the planner instead limits warming with a rising carbon price, preventing escalating climate losses and raising long-run growth relative to laissez-faire. Empirical damage estimates span both regimes, but even moderate estimates place the economy in the high-damage regime in which optimal policy limits warming and raises long-run growth.

Complementing this optimal-policy analysis, in "Does Unilateral Decarbonization Pay for Itself?" [6] (with Adrien Bilal, *American Economic Association Papers and Proceedings*, 2025), we conduct a simple benefit–cost analysis combining our global climate-damage estimates with engineering estimates of marginal abatement costs. We find that domestic benefits alone justify more than 80 percent decarbonization in the United States and European Union, even without international coordination. Our mortality estimates [16] strengthen this case further: because domestic mortality damages scale with population as well as income, unilateral mitigation can also be cost-effective for populous, lower-income economies.

Energy markets, supply chains, and trade. My second line of work studies how major global shocks—such as disruptions in energy markets, supply chains, and international trade—affect the macroeconomy. I provide direct causal evidence on these questions by exploiting institutional features and financial-market variation, as well as information from narrative and historical records.

Energy markets. Identifying the causal effects of oil shocks is challenging because oil prices are endogenous and respond simultaneously to supply, demand, and expectations. In "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements" [11] (*American Economic Review*, 2021), I make progress on this challenge by proposing a new identification strategy that exploits institutional features of OPEC meetings and high-frequency movements in oil futures to isolate news about future oil supply. Adverse news raises oil prices immediately, while production falls only gradually and inventories increase. U.S. activity contracts, inflation and inflation expectations rise, and the dollar depreciates. The results show that expectations about future supply are an important source of oil-price fluctuations.

In "From Importer to Exporter: Oil Shocks and the U.S. Economy" [2] (with James H. Stock and Luca Zanotti, *Brookings Papers on Economic Activity*, forthcoming), we ask how the shale revolution altered this transmission. A time-varying model shows that the contractionary U.S. response to adverse oil supply news weakened over time and eventually became expansionary as the petroleum trade balance improved. State-dependent local projections connect the change directly to importer-versus-exporter status. The mechanism is not a mechanical increase in aggregate net exports. Instead, the evidence points to broad-based general-equilibrium income and wealth effects: the terms-of-trade response flips from deterioration to improvement, while production, employment, consumption, and equity valuations strengthen across sectors and regions.

Another major transformation over this period has been the secular decline in the economy's reliance on energy, with energy use growing far more slowly than output. This raises the question of what has driven this striking decoupling. In "Unraveling the Drivers of Energy-Saving Technical Change" [13] (with Charles Williamson, 2025, *under revision for the Journal of Monetary Economics*), we construct a quarterly measure of U.S. energy-saving technology and use a semi-structural model to identify its underlying drivers. Higher energy prices stimulate energy-saving innovation, but they also cause an economic downturn and crowd out other innovation. Among the shocks we consider, only energy-saving technology shocks can explain the persistent divergence between output and energy use, and their effects provide little evidence of a rebound in energy consumption.

Trade policy, supply chains, and international propagation. A large literature studies the effects of tariffs using firm- and industry-level variation, but there is much less direct evidence on their aggregate macroeconomic effects. One reason is that U.S. tariffs exhibited relatively little time-series variation for much of the postwar period. In "The Macroeconomic Effects of Tariffs: Insights from 180 Years of U.S. Trade Policy" [12] (with Régis Barnichon, Tamar Den Besten, and Aayush Singh, 2026, *under revision for the Quarterly Journal of Economics*), we address this challenge by taking a long historical perspective and constructing a narrative history of U.S. trade policy from 1840 to 2024. We classify major legislative changes, multilateral negotiations, and temporary surcharges by their stated motivations and use plausibly exogenous reforms as an instrument for structural tariff shocks. Tariff increases are contractionary despite their protective intent: imports fall immediately, exports decline with a lag, and output and manufacturing activity fall persistently.

Recent pandemics, wars, and shifts in trade policy have put global supply chains under growing strain, raising the question of how disruptions to the movement of goods propagate to the broader economy. In "Supply Chain Shocks and the Macroeconomy: Evidence from Global Shipping Disruptions" [21] (with Ramya Raghavan, 2026), we isolate disruptions at maritime choke points such as the Suez and Panama Canals using narrative evidence and high-frequency financial data. These events cause a large and persistent increase in shipping costs, followed by declines in economic activity and persistent increases in producer and consumer prices. Delivery times and shortages rise, without a corresponding increase in geopolitical risk. Using granular input-output data, we also estimate macro elasticities of substitution across intermediate inputs, finding limited substitution in the short run but substantially greater reallocation across suppliers over time. The evidence provides new empirical targets for quantitative trade and production-network models.

This work is part of a broader agenda on how shocks propagate through interconnected economies. In "Growth Spillovers and Life Expectancy" [17] (with Adrien Bilal and Krzysztof Lisiecki, 2026), we study how economic growth generates health benefits across borders. We show that increases in global GDP substantially reduce mortality worldwide, with effects far exceeding those of country-specific growth. The results are robust to instrumenting global GDP with U.S. monetary policy shocks, and the evidence points to nutritional improvements in low-income food-exporting countries as an important channel. In a growth model with finite lifespans and health production, these spillovers explain over 70 percent of worldwide life-expectancy gains between 1960 and 2019, compared with around 10 percent for local growth. Internalizing these spillovers raises welfare by 6 percent.

In "International Inflation Spillovers: The Role of Different Shocks" [9] (with Gregor Bäurle and Matthias Gubler, *International Journal of Central Banking*, 2021), we show that similar movements in

foreign inflation can produce very different domestic outcomes, depending on the underlying shock. In "The Distributional Impact of the Pandemic" [10] (with Sinem Hacıoğlu-Hoke and Paolo Surico, *European Economic Review*, 2021), we use UK transaction data to show that high-income households drove much of the pandemic consumption decline, while lower-income households suffered larger earnings losses that government transfers substantially cushioned.

My ongoing research extends the narrative identification approach to another central margin of globalization. In "The Economic Effects of Immigration Policy" [28] (with Tamar Den Besten and Patrick Leitloff), we use historical U.S. data to assess whether immigration benefits or harms the labor market and the broader economy, tracing the effects of immigration policy on employment, wages, prices, and aggregate economic activity.

Heterogeneity, expectations, and policy transmission. My third line of work studies empirically and theoretically how macroeconomic policies affect households and the broader economy, with a focus on how heterogeneity, expectations, and financial frictions shape their transmission.

Inequality as a propagation mechanism. In "Capital and Income Inequality: An Aggregate-Demand Complementarity" [8] (with Florin Bilbiie and Paolo Surico, *Journal of Monetary Economics*, 2022), we identify a powerful interaction between heterogeneity in household saving and heterogeneity in income. In a tractable saver-spender model, the consumption response to aggregate-demand shocks when both forms of heterogeneity are present is much larger than the sum of their separate effects. Intuitively, investment by savers increases aggregate income, some of which accrues to spenders. Spenders consume this additional income, further boosting aggregate demand. We show that the same complementarity operates quantitatively in a heterogeneous-agent New Keynesian model with capital, idiosyncratic risk, and realistic income and wealth distributions. A robust prediction is that consumption inequality is more countercyclical than income inequality, consistent with the data.

In "Greed? Profits, Inflation, and Aggregate Demand" [14] (with Florin Bilbiie, 2024, *under revision for the American Economic Journal: Macroeconomics*), we extend this line of work to study the role of inequality and income distribution in shaping inflation dynamics. Motivated by the recent policy debate around "greedflation", we ask whether rising profits can themselves drive inflation in the New Keynesian framework. We show that in standard monetary models, procyclical profits tend to dampen rather than amplify inflation. We derive analytical conditions under which profits, inflation, and demand comove positively—reconciling the data with model predictions—and show that heterogeneity in profit income can magnify or mitigate these effects depending on the distribution of profits.

Fiscal policy and expectations. In "The Macroeconomic Effect of Stimulus Checks: Evidence from Postwar Veterans' Payments" [19] (with João Guerreiro, Jonathon Hazell, and Ed Manuel, 2026), we study a natural experiment created by large payments to U.S. veterans after World War II that mimic modern stimulus checks—one-off, lump-sum, deficit-financed transfers—but were plausibly exogenous to current and prospective economic conditions. The payments generate a temporary increase in transfers and a persistent, hump-shaped increase in aggregate consumption. The cumulative transfer multiplier reaches one after six months and exceeds two after a year.

These dynamics are difficult to reconcile with standard heterogeneous-agent New Keynesian models: while such models can generate large marginal propensities to consume on impact, they cannot reproduce the hump-shaped aggregate consumption response. A model with imperfect ex-

pectations reproduces this response, with households underreacting on impact and overreacting at longer horizons. The results therefore point to an important role for expectations in the aggregate transmission of transfers. In follow-up work on the long-run effects of transfers [29] (with João Guerreiro, Jonathon Hazell, and Ed Manuel), we use the same historical setting to study whether temporary payments have persistent effects on labor supply, wealth accumulation, and local productive capacity, providing evidence on potential fiscal hysteresis.

Moving beyond transfers, I also study how narrative identification can be used more systematically to study fiscal policy shocks. In ongoing work [27] (with Tamar den Besten and Patrick Leitloff), we develop best practices for using large language models to scale narrative identification in macroeconomics and apply them to identify tax policy shocks across advanced economies. The resulting panel allows us to estimate tax multipliers with much greater precision than traditional country-by-country narrative analyses.

My future work will continue to pursue this goal of recovering the "missing intercept" in macroeconomics: identifying the aggregate effects of major shocks and policies that cross-sectional comparisons alone cannot reveal, and using these estimates to make progress on some of the most pressing questions in macroeconomics.

Published and forthcoming papers

1. Diego R. Känzig. "The Unequal Economic Consequences of Carbon Pricing". *American Economic Review* (forthcoming).
2. Diego R. Känzig, James H. Stock, and Luca Zanotti. "From Importer to Exporter: Oil Shocks and the U.S. Economy". *Brookings Papers on Economic Activity* (forthcoming).
3. Michael D. Bauer, Diego R. Känzig, and Glenn D. Rudebusch. "Carbon Pricing and Inflation Expectations". *The Econometrics Journal* (forthcoming).
4. Saki Bigio, Diego R. Känzig, Pablo Sánchez, and Conor Walsh. "Carbon Pricing and Inequality: A Normative Perspective". *The Economic Journal* 136.677 (2026), pp. 1659–1682.
5. Adrien Bilal and Diego R. Känzig. "The Macroeconomic Impact of Climate Change: Global vs. Local Temperature". *The Quarterly Journal of Economics* 141.2 (2026), pp. 889–944.
6. Adrien Bilal and Diego R. Känzig. "Does Unilateral Decarbonization Pay for Itself?" *American Economic Association Papers and Proceedings* 115 (2025), pp. 369–373.
7. Diego R. Känzig and Maximilian Konradt. "Climate Policy and the Economy: Evidence from Europe's Carbon Pricing Initiatives". *IMF Economic Review* 72 (2024), pp. 1081–1124.
8. Florin O. Bilbiie, Diego R. Känzig, and Paolo Surico. "Capital and Income Inequality: An Aggregate-Demand Complementarity". *Journal of Monetary Economics* 126 (2022), pp. 154–169.
9. Gregor Bäurle, Matthias Gubler, and Diego R. Känzig. "International inflation spillovers: The role of different shocks". *International Journal of Central Banking* 17.1 (2021), pp. 191–230.
10. Sinem Hacıoğlu-Hoke, Diego R. Känzig, and Paolo Surico. "The distributional impact of the pandemic". *European Economic Review* 134 (2021), p. 103680.
11. Diego R. Känzig. "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements". *American Economic Review* 111.4 (2021), pp. 1092–1125.

Under revision

12. Régis Barnichon, Tamar Den Besten, Diego R. Känzig, and Aayush Singh. "The Macroeconomic Effects of Tariffs: Insights from 180 Years of U.S. Trade Policy". 2026. *Reject and resubmit, The Quarterly Journal of Economics*.
13. Diego R. Känzig and Charles Williamson. "Unraveling the Drivers of Energy-Saving Technical Change". 2025. *Revise and resubmit, Journal of Monetary Economics*.
14. Florin O. Bilbiie and Diego R. Känzig. "Greed? Profits, Inflation, and Aggregate Demand". 2024. *Revise and resubmit, American Economic Journal: Macroeconomics*.

Working papers

15. Adrien Bilal, Thibault Ingrand, and Diego R. Känzig. "Balancing Growth in a Warming World". 2026.
16. Adrien Bilal, Diego R. Känzig, and Krzysztof Lisiecki. "Global Temperature and Global Mortality". 2026.
17. Adrien Bilal, Diego R. Känzig, and Krzysztof Lisiecki. "Growth Spillovers and Life Expectancy". 2026.
18. Konstantinos Gavriilidis, Diego R. Känzig, Ramya Raghavan, and James H. Stock. "The Macroeconomic Effects of Climate Policy Uncertainty". 2026.

19. João Guerreiro, Jonathon Hazell, Diego R. Känzig, and Ed Manuel. "The Macroeconomic Effect of Stimulus Checks: Evidence from Postwar Veterans' Payments". 2026.
20. Diego R. Känzig, Maximilian Konradt, Lixing Wang, and Donghai Zhang. "Innovation, Business Cycles, and the Climate Transition". 2026.
21. Diego R. Känzig and Ramya Raghavan. "Supply Chain Shocks and the Macroeconomy: Evidence from Global Shipping Disruptions". 2026.
22. Diego R. Känzig, Julian Marenz, and Marcel Olbert. "Carbon leakage to developing countries". 2024.

Work in progress

23. Lint Barrage and Diego R. Känzig. "Green Fiscal Multipliers".
24. Adrien Bilal, Emirhan Ilhan, Diego R. Känzig, Karsten Müller, and Chenzi Xu. "Climate Change and Financial Stability: 1850–2024".
25. Adrien Bilal and Diego R. Känzig. "Climate Damages Across the Globe".
26. Adrien Bilal and Diego R. Känzig. "Prices and Quantities in a Warming World".
27. Tamar Den Besten, Diego R. Känzig, and Patrick Leitloff. "Large Language Models for Narrative Identification: Tax Multipliers Across Countries".
28. Tamar Den Besten, Diego R. Känzig, and Patrick Leitloff. "The Economic Effects of Immigration Policy".
29. João Guerreiro, Jonathon Hazell, Diego R. Känzig, and Ed Manuel. "The Long-Run Effects of Transfers".